Factory Machines

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1620 | Sorting and Searching | https://cses.fi/problemset/task/1620

A factory has n machines which can be used to make products. Your goal is to make a total of t products.

For each machine, you know the number of seconds it needs to make a single product. The machines can work simultaneously, and you can freely decide their schedule.

What is the shortest time needed to make t products?

Input

The first input line has two integers n and t: the number of machines and products.

The next line has n integers $k_1,k_2,\ldots,k_n$: the time needed to make a product using each machine.

Output

Print one integer: the minimum time needed to make t products.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le t \le 10^9$
- $1 \le k_i \le 10^9$

Example

Input
3 7 3 2 5

Output
8

Explanation: Machine 1 makes two products, machine 2 makes four products and machine 3 makes one product.